Page 1 | H7-A: Pilot evidence and model assumptions

GreenWaste is a fictional training case. Use this summary to check every claim made in the session.

§1 | The pilot RCT result

A lottery assigned **200 eligible businesses** in one pilot district to GreenWaste and **200** to a control group. Annual waste management cost was measured at 12 months. The treatment-minus-control estimate is **-1,014 AED** per business (95% interval **-1,167 to -862 AED**). On a saving scale, the estimate is **1,014 AED per business per year**, with a 95% interval from **862 to 1,167 AED**. Assignment supports a causal interpretation in the pilot if implementation and measurement remain credible. It does not establish effects in other districts or over five years.

§2 | Which economic question?

Method	Outcome unit	A question it can answer
Cost-effectiveness analysis (CEA)	Natural unit	AED per tonne of landfill avoided
Cost-utility analysis (CUA)	Health utility, often QALYs	AED per QALY gained
Cost-benefit analysis (CBA)	Money	Net benefit in AED or benefit-cost ratio

CUA is a form of CEA. The pilot's measured waste-cost outcome is in AED and can be considered as a monetary benefit, subject to a clear perspective and what the bill reduction represents. This handout works through a **partial monetary model**, not a complete social CBA. It has no causal landfill-effect or QALY calculation for CEA or CUA.

§3 | Inputs to the worked model

Table I: Per pilot business

Input	Value	Source
Annual saving per participant business	1,014 AED	Pilot RCT estimate at 12 months
One-off cost per participant business	1,800 AED	Assumption in this fictional case
Year when savings start	Year 1	Assumption
Number of years that savings last	5 years	Assumption
Discount rate	5% a year	Assumption

The model treats the RCT's 12-month annual-cost difference as a **year-1 saving**, then assumes that the same saving persists in **years 2 to 5**. It assumes **1,800 AED** in delivery cost at year 0 and a **5%** annual discount rate. Only the 12-month difference is estimated from the RCT. The cost, persistence and discount rate are teaching assumptions, not audited or official inputs. Calculations use unrounded values; displayed amounts are rounded.

§4 | Whose costs and benefits?

The assumed cost is paid by the **authority**. The measured financial saving is for **businesses**. Staff time, business maintenance and wider environmental effects have not been valued. A lower bill may reflect saved resources or a transfer between actors. This ratio is neither an authority budget return nor a complete social benefit-cost analysis. Specify the perspective and check what is already counted before adding entries.

Method references: CDC, Cost-Effectiveness Analysis; NICE, Health Technology Evaluations Manual; HM Treasury, The Green Book (2026). The fictional model does not claim to follow an official discount rate.

Page 2 | H7-A: Timing, results and sensitivity

§5 | Cash flow per pilot business

Table F: AED per business; each saving shown at its year-0 present value

Year	Cost (AED)	Saving (AED)	Saving at value today (AED)
0	1,800	0	0
1	0	1,014	966
2	0	1,014	920
3	0	1,014	876
4	0	1,014	835
5	0	1,014	795

For year 1, the value today of the annual saving is **1,014 / 1.05 = 966 AED**. The savings in years 2 to 5 are divided by 1.05 raised to their respective year numbers. Their persistence is assumed, not observed in the RCT.

Table O: Worked model, per pilot business

Measure	Value	What it is
Benefits at value today	4,392 AED	The modelled savings in years 1 to 5, converted to year 0
Cost at value today	1,800 AED	The one-off equipment delivery cost at year 0
Benefit-cost ratio	2.44	Benefits divided by costs
Net present value	2,592 AED	Benefits minus costs

The benefit-cost ratio is present-value benefits divided by present-value costs; it has **no units**. Net present benefit is the difference, in **AED**. Under this model, an annual saving of about **416 AED** gives a ratio of 1. That model break-even level differs from the separate policy rule of at least **1,000 AED annual waste-cost saving**. The RCT interval crosses the policy rule. It is not a confidence interval for the benefit-cost ratio.

§6 | Change the assumption, then check the result

Scenario	Modelled ratio
Base model	2.44
Savings fade by 30% after the first saving year	1.40
Add 600 AED to year-0 delivery cost	1.83
Discount at 8% instead of 5%	2.25
Savings last three years rather than five	1.53
Savings fade by 30% and add 800 AED cost	0.97

These are hypothetical settings, not findings about GreenWaste. The final scenario changes two inputs. Their ranking depends on the values chosen. Programme accounts, follow-up data and applicable appraisal guidance would help determine realistic inputs.

Page 3 | Work the economic question

Q1 | Separate evidence from assumptions (H7-A §1 and Table I)

a. What outcome did the pilot RCT estimate? Include unit, period and population.

b. Where does the 1,800 AED delivery cost come from? What record would verify it?

c. Which modelled years have no RCT follow-up evidence?

Q2 | Choose the unit (H7-A §2)

a. To calculate AED per tonne of landfill avoided, which effect and cost would you need?

b. What is the outcome unit in CUA? Why can we not calculate it here?

c. Which method expresses benefits and costs in AED? Why is our ratio only partial?

Q3 | Choose a perspective (H7-A §4)

a. Who bears the included cost? Who receives the included financial saving?

b. Name one missing item and the source that could establish its amount and timing.

Q4 | Read the arithmetic (Tables F and O)

a. The year-1 saving at its value today is 1,014 / _____ ≈ _____ AED.

b. Present-value benefits are _____ AED; assumed delivery cost is _____ AED.

c. Show the benefit-cost ratio: _____. Show net present benefit: _____.

d. Why does multiplying 1,014 by five fail to give the model's present value?

Page 4 | Question the result and advise

Q5 | Two thresholds (H7-A §5)

The model breaks even at about **416 AED** annual saving. The policy minimum is **1,000 AED** annual saving.

a. Does the pilot RCT interval of 862 to 1,167 AED sit entirely above either threshold? Explain the difference.

b. Name an assumption that could move the model break-even level.

Q6 | Scenario and evidence (H7-A §6)

a. Predict which single-input scenario will lower the ratio most. Then check the table: was your prediction right?

b. What does the combined scenario of fading savings and 800 AED extra cost show? What does it **not** show?

c. Request one specific record that would help you choose a more realistic model input.

Q7 | Correct the claim

“The RCT proves GreenWaste saves 1,014 AED every year for five years. Its benefit-cost ratio of 2.44 proves citywide expansion is good value for money.”

Underline what the RCT supports. Cross out the unsupported steps. Write a corrected sentence.

Q8 | Your provisional advice

State what the pilot-based model suggests **under its assumptions**. Then give a provisional action and one evidence request that could change it.